A four-card check of the printed shelf-shuffle spectrum

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Abstract

The printed nonzero-spectrum conjecture for a single-shelf shuffle includes the value 1/16 when the deck has four cards. We prove that this value is not an eigenvalue of the four-card position matrix. A Lean certificate checks both the matrix obtained by enumerating all fair-coin choices and its agreement with the source's explicit binomial formula, then excludes 1/16 over the complex field. This is an index-range correction to the conjecture as printed, not a claim of a general new spectral formula.

1 The printed claim

In the single-shelf model of Clay [1], cards initially labeled $1, \dots, n$ are drawn from the bottom of the deck. Each drawn card is placed on the top or bottom of the current pile, with equal probability. The position matrix has entries

$$M_{ij} = \frac{\binom{i-1}{j-1} + \binom{i-1}{n-j}}{2^i} \quad (1 \le i, j \le n).$$

Conjecture 2 of the cited version, also MathDB369281 [2], lists the nonzero eigenvalues as

$$2^{-2i}, \qquad 0 \le i \le \left\lceil \frac{n-1}{2} \right\rceil.$$

For $n = 4$, the stated upper endpoint is 2, so the list includes 1/16.

Theorem 1. *For a four-card deck,* 1/16 *is not an eigenvalue of the position matrix. Consequently the nonzero-spectrum conjecture is false as printed.*

Proof. Substitution into the binomial formula gives

$$M = \begin{pmatrix} 1/2 & 0 & 0 & 1/2 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/8 & 3/8 & 3/8 & 1/8 \\ 1/8 & 3/8 & 3/8 & 1/8 \end{pmatrix}.$$

Suppose $Mv = v/16$ for a complex column vector $v = (a, b, c, d)^T$. Multiplying by 16 gives

$$\begin{aligned} 7a + 8d &= 0, \\ 4a + 3b + 4c + 4d &= 0, \\ 2a + 6b + 5c + 2d &= 0, \\ 2a + 6b + 6c + d &= 0. \end{aligned}$$

Subtracting the third equation from the fourth gives $c = d$. Twice the second equation minus the fourth therefore gives $6a + 9d = 0$. Together with $7a + 8d = 0$, this implies $5d = 0$, then $a = b = c = d = 0$. There is no nonzero eigenvector, proving the theorem. □

2 Verification of the shuffle model

The certificate does not merely postulate the displayed numerical matrix. It enumerates all $2^4 = 16$ Boolean choices for top-versus-bottom placement, processes cards $4, 3, 2, 1$, and counts the final position of each card. The resulting count matrix is

$$\begin{pmatrix} 8 & 0 & 0 & 8 \\ 4 & 4 & 4 & 4 \\ 2 & 6 & 6 & 2 \\ 2 & 6 & 6 & 2 \end{pmatrix}.$$

The first placement coin has no effect on an empty pile; retaining it double-counts every effective outcome equally, which preserves the uniform distribution. Division by 16 yields M exactly. Lean checks this finite enumeration and its agreement with the source formula by kernel reduction, without a native decision axiom.

The file `formal/ShuffleSpectrum.lean` then proves that any complex vector satisfying $Mv = v/16$ is zero. It expresses the missing eigenvalue using mathlib's `Module.End.HasEigenvalue`, and explicitly negates the universal printed spectral claim using $n = 4$ and $i = 2$. The proof includes the nonzero-eigenvector condition and no unproved spectral premise.

3 Scope

This note targets the ceiling endpoint in the version cited. It does not disprove a conjecture with a different endpoint, claim to determine the complete spectrum for arbitrary deck size, or claim that this correction was previously unknown. In particular, changing the upper endpoint is a different statement and requires its own proof. The certificate is a complete refutation of the printed statement, with a transparent four-card example. The neighboring quadratic-eigenvector conjecture is a separate question.

The implementation uses Lean 4.34.1 and the campaign's pinned mathlib revision. All local declarations are audited for standard logical axioms only. This is an AI-assisted verification note, not external peer review. Original prose is CC BY 4.0 and original proof code is MIT.

References

[1] A. Clay, *Guessing Strategies for Shelf-Shuffling Machines*, arXiv:2507.10294v1, July 14, 2025. Theorem 1.1 and Conjecture 2. https://arxiv.org/html/2507.10294v1.

[2] MathDB, *The conjectured nonzero spectrum of the position matrix*, problem 369281. Accessed October 25, 2026. https://mathdb.com/p/369281.